

To find the maximum possible value of g , we are given that g divides $(P/a_i) + 1$ for each $i = 1, \dots, k$, where $P = \prod_{j=1}^k a_j$.

This divisibility condition can be written as:

$$\frac{P}{a_i} \equiv -1 \pmod{g}$$

Multiplying both sides by a_i , we get:

$$P \equiv -a_i \pmod{g}$$

Since P is a constant for all i , this implies that all the a_i must be congruent to the same value modulo g . Because the a_i are distinct positive integers, we can order them such that $a_1 < a_2 < \dots < a_k$, and they must be of the form:

$$a_i = a + c_i g$$

for some initial positive integer remainder a (with $1 \leq a \leq g - 1$) and integers $0 = c_1 < c_2 < \dots < c_k$.

To minimize the sum of the sequence relative to g and thereby allow g to be as large as possible without exceeding the maximum sum, we should pick the smallest possible number of elements, $k = 3$, and the smallest possible step sizes, $c_2 = 1$ and $c_3 = 2$. Thus, our integers are $a_1 = a$, $a_2 = a + g$, and $a_3 = a + 2g$.

The sum of these three numbers is strictly less than $3 \cdot 10^6$:

$$a_1 + a_2 + a_3 = 3a + 3g < 3 \cdot 10^6 \implies a + g \leq 999999$$

We are given g divides $(P/a_1) + 1 = a_2 a_3 + 1$, which gives:

$$(a + g)(a + 2g) + 1 \equiv 0 \pmod{g} \implies a^2 + 1 \equiv 0 \pmod{g}$$

Thus, $a^2 + 1$ must be a multiple of g . Let $a^2 + 1 = mg$ for some positive integer m . We want to maximize $g = \frac{a^2 + 1}{m}$ subject to the condition $a + g \leq 999999$. Let's test the smallest values for m (since g roughly scales inversely with m):

1. **For** $m = 1$: $a^2 + 1 = g$. The boundary condition requires $a + a^2 + 1 \leq 999999 \implies a^2 + a - 999998 \leq 0$. The largest integer satisfying this is $a = 999$. This yields $g = 999^2 + 1 = 998002$.
2. **For** $m = 2$: $\frac{a^2 + 1}{2} = g$. The boundary condition requires $a + \frac{a^2 + 1}{2} \leq 999999 \implies a^2 + 2a + 1 \leq 1999999 \implies (a + 1)^2 \leq 1999999$. The largest integer satisfying this is $a + 1 = 1414 \implies a = 1413$. Because 1413 is odd, $1413^2 + 1$ is even, making it perfectly divisible by 2. This yields $g = \frac{1413^2 + 1}{2} = \frac{1996570}{2} = 998285$.
3. **For** $m \geq 3$: As m continues to rise, the bounds strictly limit g to smaller values. (For $m = 5$, the maximum g is 997258; $m = 3, 4$ have no modular solutions). By setting bounding continuous limits, we can prove g mathematically cannot exceed roughly 998268 for all $m \geq 3$.

Thus, the global maximum occurs when $m = 2$, yielding $g = 998285$ generated by the terms: $a_1 = 1413$ $a_2 = 999698$ $a_3 = 1997983$

Their sum is $1413 + 999698 + 1997983 = 2999094 < 3 \cdot 10^6$. Their respective values of $(P/a_i) + 1$ are beautifully divided by 998285 giving the exact integers 2000811, 2828, and 1415 respectively.

Final Answer: The maximum possible value of g is **998285**.